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Modeling and Monitoring of Within-Match Fatigue in Professional Football

by

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Dissertation Presented for the Degree of
MSc in Statistics with Data Science

June 30, 2026

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Executive Summary

Using GPS and accelerometer data collected from every player during the 2025/26 Scottish Premiership season, we developed a system that tracks how physically fatigued each player is during the second half of a match. We measure how much each player's explosive output (their high-intensity accelerations and decelerations) have dropped off compared to their own peak performance in the first half of the match. We found that players substituted out during matches showed significantly higher fatigue levels than those who stayed on the pitch. We also found that fatigue does not affect all players equally, differing between player position, players with less recovery time between matches, if they are playing at home, etc.

The system produces two practical outputs for match day use. A real-time insight which ranks all starting players by how fatigued they currently are, giving coaching staff a glance at who is struggling the most physically. A half-time report that identifies player who showed the highest fatigue at the end of the first half, giving the coaching staff a heads up on who may need some managing in the second half.

An honest limitation of our system is that it cannot reliably predict exactly how fatigued a specific player will be in five or ten minutes time. Fatigue in a live match is influenced by a lot of factors including tactical decisions, the opponent, psychological state, etc, that the GPS data alone cannot capture. What the system can do reliably is identify and flag which players are most at risk.

Importantly, this system does not make substitution decisions, that remains with the coaching staff, who bring tactical knowledge and experience that no current data system can replace. What it does do is provide objective physical evidence so that the decision about who to substitute and when is informed by how each player is holding up.

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AI disclosure

Generative AI was only used to assist in *minor* code edits or errors.

Word Count: 4950

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1 Introduction

Physical fatigue is a fundamental aspect of performance in professional football. As a match progresses, players experience a decline in their capacity to produce high-intensity explosive actions. This can have consequences for team performance where fatigued players are at higher risk for injury and decreased performance outputs [8]. The challenge for coaching staff, particularly when making substitution decisions, is managing this player fatigue throughout the match.

GPS and accelerometer data from Hibernian FC’s 2025/26 Scottish Premiership season can potentially support this challenge by providing objective physical evidence, including real-time data on high-intensity accelerations, high-intensity decelerations, high speed running, maximum velocity, and total distance covered. The data covers 35 matches across 28 players with analyses restricted to starting players. Full details on the data structure and preprocessing are provided in Appendix A. The process of translating this data into actionable monitoring outputs remains a challenge that this paper attempts to address.

The goal of this research is to develop a statistical method to monitor real-time within-match fatigue to provide coaching staff with evidence to support decision making instead of making explicit substitution recommendations. The central question addressed in this research is: can we reliably characterize, monitor, and predict within-match player fatigue? The player measure of within-match fatigue was defined as the decline in a players rolling high-intensity acceleration and deceleration output relative to their own first half peak.

The paper is structured as follows. Section 2 defines the fatigue and deterioration measures and presents the exploratory analysis and supporting literature that validates their use. Section 3 describes the model specifications and evaluation. Section 4 presents the model comparison results and diagnostics. Section 5 describes the real-time monitoring applications and outputs. Section 6 discusses the key findings, acknowledges limitations, and identifies potential future work.

2 Fatigue Operationalization

2.1 Fatigue Definition

A common definition of fatigue by Edwards states that fatigue is the "failure to maintain the required or expected force (or power output)" [3]. For the purpose of our research into fatigue of Hibernian FC football players, we build off this definition by pulling from relevant data context. The data, as detailed in Appendix A, is made up of only external load metrics, defined as the physical work performed by the athlete [7]. This includes tracking the metrics total distance covered (m), high speed distance covered (m), max velocity (m/s^2), high-intensity accelerations (count), and high-intensity decelerations (count). This differs from internal load which is the physiological and psychological stress imposed upon an athlete, which may be measured by perceived exertion, heart rate, or blood lactate concentrations [7]. Both load types have been found to be accurate fatigue indicators in athletes [7], however, given we only have external load data, we propose a fatigue definition based on performance degradation.

2.2 Exploratory Analysis of Fatigue Metrics

To evaluate which external load metrics were most sensitive to fatigue, 10 minute right rolling windows were computed for each of the five available metrics and plotted across match time for all player-match combinations in the training set. A window of 10 minutes was selected to sufficiently smooth minute-to-minute variation while keeping it narrow enough to capture genuine within-match deterioration. As shown in Figure 1, high-intensity accelerations (HIA) and high-intensity decelerations (HID) have particularly evident declines across a match. High speed distance sees some decline while max velocity and distance covered don’t show significant declines.

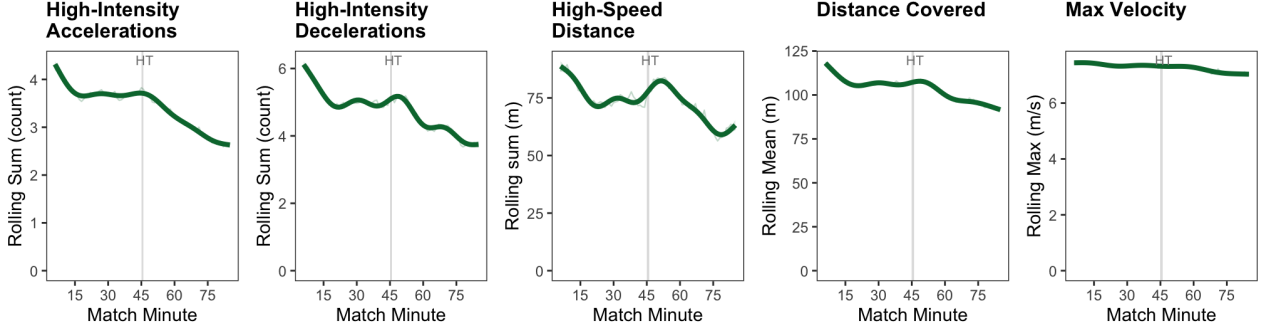


Figure 1: Rolling trajectories across match time all five external load metrics.

Previous sports science literature supports these findings by highlighting the particular effect that high-intensity actions have on football players. High-intensity actions, accelerations and decelerations in particular, were found to have higher exertion costs than other external metrics, with associations to reduced neural drive and muscle damage markers [2, 8, 13]. For instance, high-intensity decelerations requires muscle contractions in the hip, knee, and ankles to generate braking forces with much greater mechanical load per meter than any other external load during match play [5, 8, 11]. Additionally, reduced accelerations and decelerations prevent sharp changes in maneuvers made by players which in turn reduce impairs tactical execution and injury risk and in turn match performance [8].

While high speed distance (HSD) covered can also be found in supporting literature as a fatigue indicator due to explosive and cardiovascular effort [12], we felt that these actions are particularly sensitive to match context. Team tactics of the home and opposing team and defensive structure greatly influences observed movement patterns making tactical contamination a challenge for speed based metrics. Thus, we chose to exclude high speed distance covered from our fatigue measure. Similarly, total distance covered was also excluded from the fatigue definition, with the compensation phenomenon in mind that suggests athletes maintain total movement volume while selectively reducing high-intensity efforts as fatigue develops [1, 14].

Thus, for the following modeling we define fatigue as a decline in the frequency of high-intensity efforts, specifically accelerations and decelerations.

2.3 Measuring Fatigue and Defining Performance Deterioration

As worked in the exploratory data analysis, to characterize the definition of HIA and HID events as fatigue indicators and to produce a more stable representation of a players explosive output, a rolling 10 minute window aggregation was used. For each player-match combination, the fatigue score at minute t was defined as the sum of high-intensity accelerations and high-intensity decelerations within a right aligned 10 minute window. We define fatigue score as,

$$\text{Fatigue Score}_t = \sum_{k=t-9}^t (\text{HIA}_k + \text{HID}_k) \quad (2.1)$$

where HIA_k and HID_k are the high-intensity acceleration and deceleration counts at minute k . This is the defined *fatigue score*, which indicates how well a player is performing based on our fatigue definition. We define *peak performance* as the largest sum of HIA and HID events in a match minute from the first half. To account for possible outliers of intense action moments, we take the 90th percentile instead of the maximum. We define peak performance as,

$$\text{Peak Performance}_i = Q_{90}(\{\text{Fatigue Score}_t : t \in \text{H1}_i\}) \quad (2.2)$$

where $Q_{90}(\cdot)$ denotes the 90th percentile, H1_i denotes the set of first half minutes for player i , and Fatigue Score_t is as defined in Equation 2.1. Peak performance is computed once per player-match

combination at half-time and held fixed for all second half minutes.

$$\text{Deterioration}_t = \frac{\text{Peak Performance}_i - \text{Fatigue Score}_t}{\text{Peak Performance}_i + \epsilon} \quad (2.3)$$

With this, *deterioration* is defined in Equation 2.3 as the proportional decline from this peak performance from Equation 2.2 to minute t in the match where $\epsilon = 0.01$ is a small constant added to the denominator to prevent division by zero for players with a peak of zero. This deterioration is our model output and is used as the indicator for our fatigue monitoring and predictions. Note that, in using first half statistics for the definition of deterioration, training and testing of models will only be completed using second half data, further details with literature support on this decision can be found in Section 3.1.

2.4 Deterioration Verification

If starter players who are substituted exhibit higher deterioration then those who remain on the pitch for the entire match, then predicting deterioration provides a basis for coaching staff to identify fatigued players and in turn potential substitution candidates. In Table 1 we examine this relationship across 12 deterioration thresholds ranging from 10% to 95% below a player’s first half peak. Substituted players consistently exceeded the threshold at a higher rate than players who played the full match. This was particularly evident at the 50% threshold, where 75.2% of substituted players had declined to 50% of their first half peak output compared to only 30.4% of full-time players having declined to 50% of their first half peak output. Similarly we find in Figure 2 that the distribution density of deterioration from substituted players has a clear right shift relative to non-substituted players indicating higher deterioration levels at the point of substitution. These findings support deterioration as a fatigue monitoring signal as players selected for substitution were not only more likely to be deteriorating, but were deteriorating at substantially greater magnitudes than their teammates who continued playing for the full match.

Deterioration Threshold	Substituted players above threshold (%)	Full time players above threshold (%)	Difference (Sub - Full)
10%	96.0%	79.4%	+16.6
20%	94.4%	69.5%	+24.9
30%	92.0%	59.0%	+33.0
40%	88.0%	44.7%	+43.3
50%	75.2%	30.4%	+44.8
60%	60.0%	19.9%	+40.1
70%	47.2%	9.8%	+37.4
75%	35.2%	6.3%	+28.9
80%	31.2%	4.1%	+27.1
85%	22.4%	2.5%	+19.9
90%	14.4%	1.4%	+13.0
95%	8.8%	0.8%	+8.0

Table 1: Percent of substituted and non-substituted players that were above deterioration thresholds

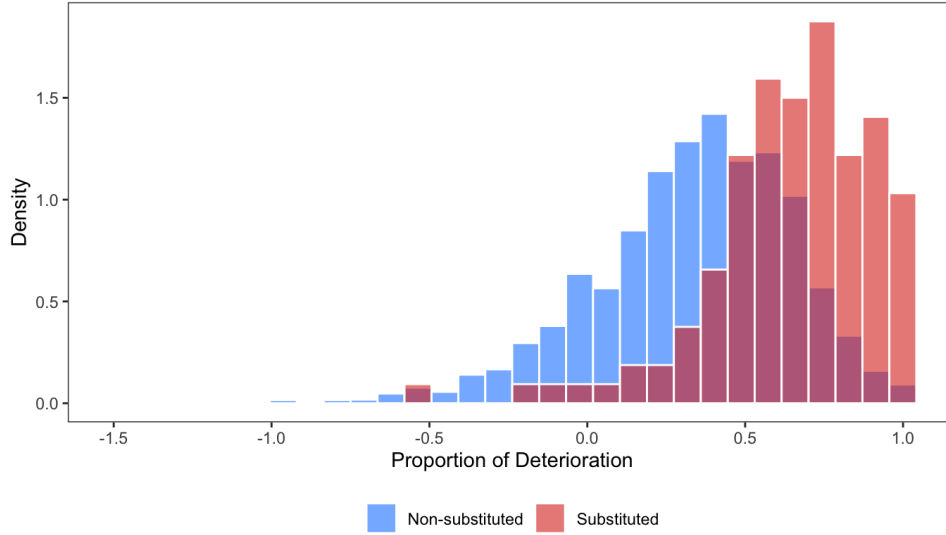


Figure 2: Density distributions of deterioration from substituted players at the minute of substitution and non-substituted full time players.

3 Methodology

Model investigation followed a simple to complex progression. Note the response variable for all models run is the deterioration rate as defined in 2.3.

3.1 Second Half Focus

By the working definition of deterioration, we will only be using second half data for modeling and predicting. While we acknowledge this cuts our training and testing data in half, we do not find this to be a detrimental limitation. Substitutions in football are informed by tactical decision making and/or to offset fatigue by pulling fresh legs into the match. Earlier data analysis in Section 2.1 and other match analyses in supporting research find high-intensity movements decline in the second half [9, 4, 10] and substitutions primarily happen at half-time or during the second half [6]. As discussed in Section 2.4, substituted players see higher levels of deterioration. This aligns with industry partners at Hibernian FC who work with the assumption that first half substitutions found in the data were likely due to injury or tactical decisions.

3.2 Exploratory Data Analysis of Predictors

To establish empirical motivation for what predictors to include in the models, we are identifying variables that show a plausible relationship with deterioration. Note all exploratory analysis is done using only second half observations, consistent with deterioration defined in Section 2.1.

Figure 3 shows the distribution of deterioration across second half minutes alongside a GAM smooth of the overall trend. Deterioration increases progressively across the second half, confirming players decline relative to their first half peak and the match progress. The non-linear shape of the smooth line provides preliminary evidence that a non-parametric representation of this time trend may be better fit.

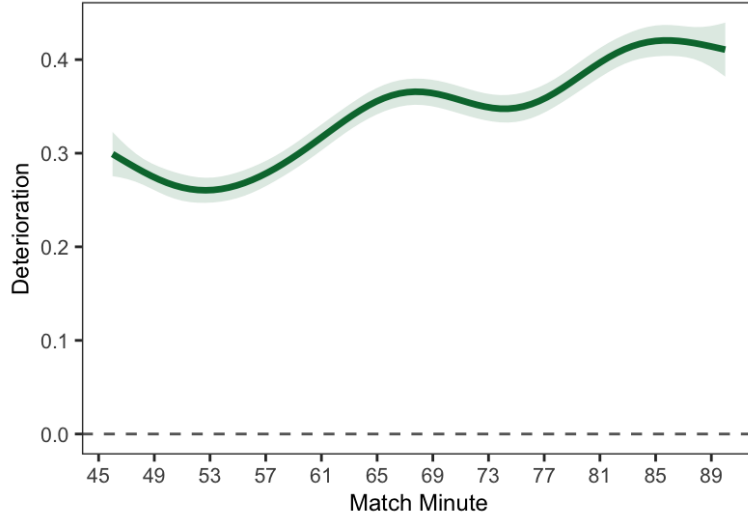


Figure 3: Deterioration across second half match minutes for all player-minute observations.

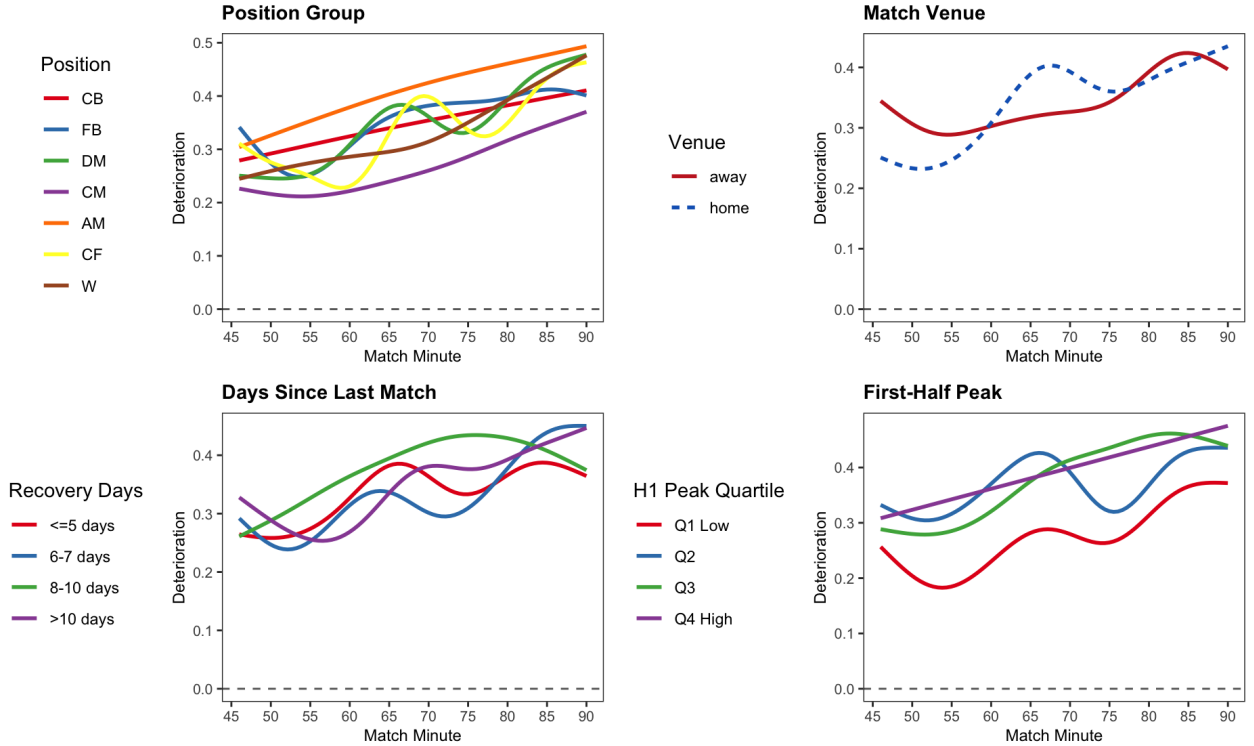
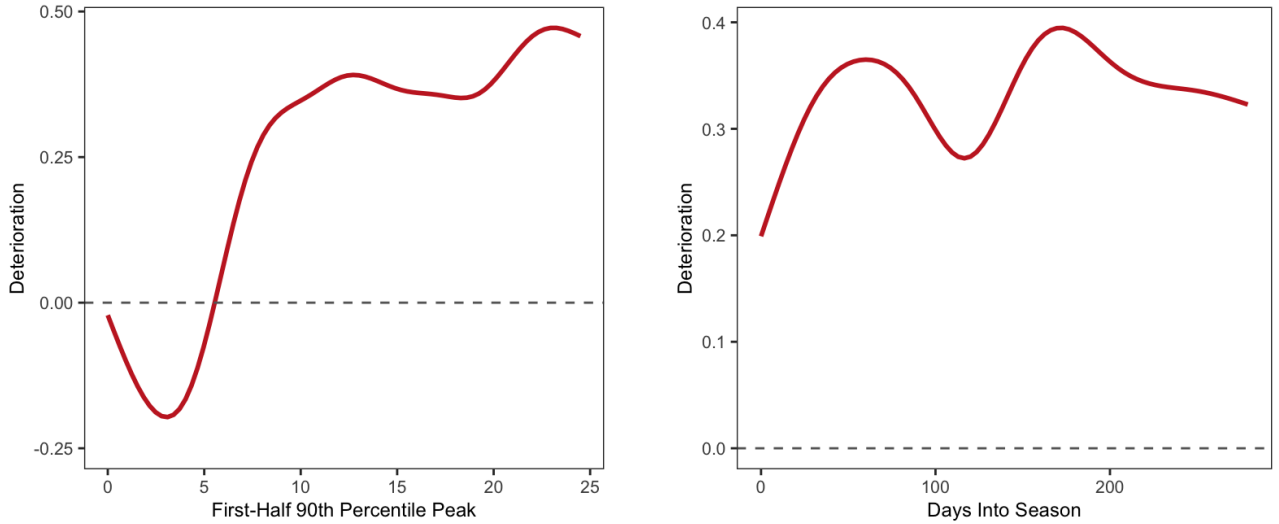


Figure 4: Deterioration trajectories across second half minutes stratified by player position, match venue, players days since last match, and and first half peak quartiles.

We then consider contextual predictors that may effect this temporal relationship. Figure 4 shows deterioration trajectories across second half minutes split by position group, match venue, days since last match, and first half peak performance. Across all four, trajectories diverge between the subgroups as the second half progresses, suggesting that the rate and shape of deterioration is potentially effected by these factors. For instance, while all positions see an increase in deterioration across the second half, center forwards show a steeper and less linear trajectory compared to center backs who had a more gradual incline. Home games shows a slightly steeper and less linear curve than away games. While days since last match don't exhibit the most explicit divergence, in conversation with industry support from Hibernian FC, there was considerable interest in how recovery days may effect players performance. With that, it was included as a predictor. Players with higher first half peaks (third

and fourth quartile) showed steeper and higher deterioration across the second half compared to those with lower peaks. Overall, the visual evidence provides motivation to include the match minute interaction terms and corresponding individual terms in our models. Note that we did consider game state (winning, drawing, or losing) as a potential predictor however plots of behavior over the second half related to deterioration as well as match minute interaction showed little relationship. Additionally, it raises the limitation of hindering models ability to accurately predict literally across match minutes as we won't know the future game state, therefore we chose not to include game state or any of its interactions in our models.



(a) Deterioration plotted against first half 90th percentile peak fatigue score.

(b) Deterioration against days into the season.

Figure 5: Relationship between deterioration and baseline predictors. Left: first half peak fatigue. Right: seasonal progression.

Figure 5 considers individual baselines of the first half peaks as well as days into season against deterioration. There is a clear positive relationship for first half peak where players with higher first half peaks tend to show more deterioration in the second half. There is a visible seasonal trend with increasing non-linear deterioration as the season progresses. This justified the inclusions as main effect predictors.

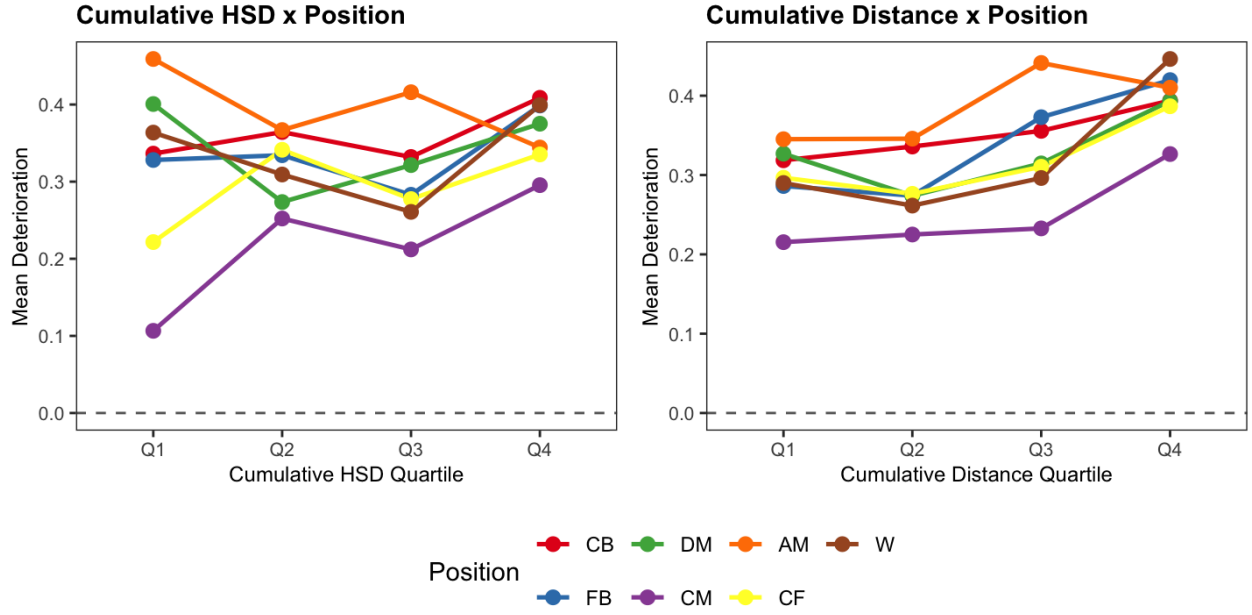


Figure 6: Mean deterioration against Cumulative High-Speed distance covered (left) and cumulative distance covered (right) quartiles.

There exists a positive relationship between cumulative high speed distance accumulated up to minute t and mean deterioration that differs between player position. A similar observation can be made of cumulative distance. This motivates our inclusion of cumulative HSD and distance covered as positional interactions in our model.

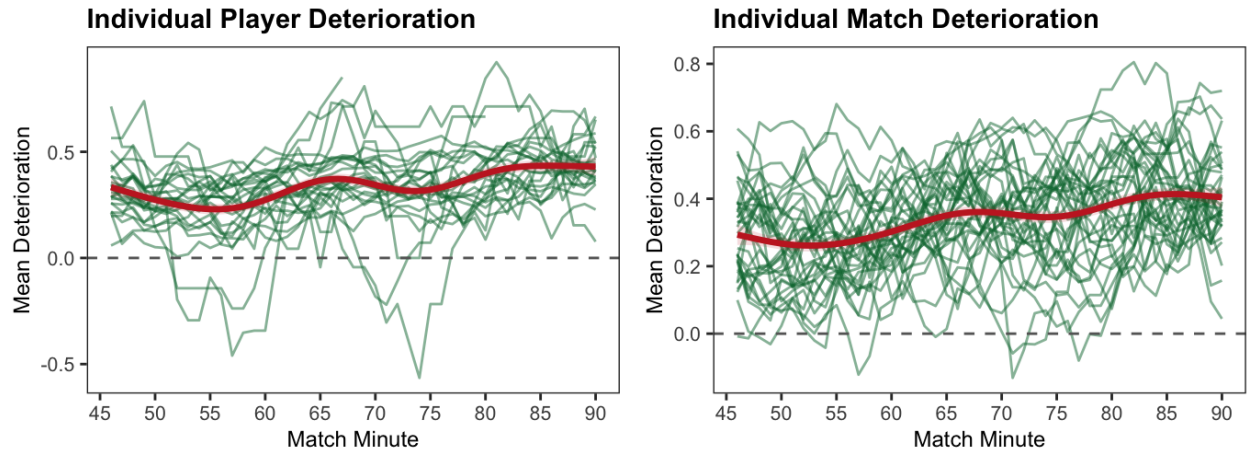


Figure 7: Match level (right) and individual player (left) mean deterioration trajectories across second half. Each line represents one match (right) or player (left) with the red curve being the overall GAM trend.

Figure 7 shows individual player and individual match deterioration trajectory across the second half. There exists a substantial spread between individual lines with some players consistently deteriorating more rapidly across matches than others and some matches showing consistently higher deterioration across players than others. This reflects the differences in player and match variation due to factors such as tactical context not visible in fixed effects. Therefore, these random effects are included in our model.

3.3 Lag Term Inclusion

We found it theoretically and operationally important to include a lagged deterioration term in our models. In real-time monitoring, only information up to minute t is available so predicting deterioration at $t + \ell$ calls for current and historical observations as predictors. This also reflects how coaching staff assess players, by observing trajectory of play rather than a single moment. This will allow our models to distinguish between sustained deterioration from temporary fatigue dips. Lags of one, five, and ten minutes were investigated. We note that, due to the rolling window construction, shorter lag terms share observations with the current fatigue score, which may slightly inflate predictive performance. Results for $\ell \in \{1, 5, 10\}$ are presented in Section 4.

3.4 Linear and Polynomial Models

Three parametric mixed-effects models were fit as baseline specifications against which the GAM approach could be compared. The progression from linear to cubic was motivated by the deterioration trajectory observed in the exploratory analysis in Section 2.2 which suggests a constant rate may be restrictive. Each model relaxes the assumption of the previous one by allowing a more flexible trajectory. The linear model is the simplest specification and assumes that deterioration evolves at a constant rate across the second half. Considering all the predictors discussed in Section 3.2, this linear mixed-effect model takes the form:

$$\begin{aligned} \text{Deterioration}_{i,t+\ell} = & \beta_0 + \beta_1 \text{Deterioration}_{i,t} + \beta_2 t + \beta_3 \text{DaysSince}_i + \beta_4 \text{DaysIntoSeason}_m \\ & + \beta_5 \text{CumDist}_{i,t} + \beta_6 \text{CumHSD}_{i,t} + \beta_7 \text{Peak}_i + \beta_8 \text{Position}_i + \beta_9 \text{Venue}_m \\ & + \beta_{10} (t \times \text{Position}_i) + \beta_{11} (t \times \text{Venue}_m) + \beta_{12} (t \times \text{DaysSince}_i) + \beta_{13} (t \times \text{Peak}_i) \\ & + \beta_{14} (\text{CumDist}_{i,t} \times \text{Position}_i) + \beta_{15} (\text{CumHSD}_{i,t} \times \text{Position}_i) \\ & + u_i + v_m + \epsilon_{i,t} \end{aligned} \quad (3.1)$$

where $\text{Deterioration}_{i,t}$ is the lagged deterioration for player i at minute t , t is match minute, DaysSince_i is the number of days since player i 's last match appearance, DaysIntoSeason_m is days elapsed since the first match of the season for match m , $\text{CumDist}_{i,t}$ and $\text{CumHSD}_{i,t}$ are cumulative within-match distance and high speed distance for player i up to minute t , Peak_i is player i 's peak fatigue score, Position_i is player position, Venue_m is binary for home or away, u_i and v_m are player and match random intercepts respectively, and $\epsilon_{i,t}$ is the residual error term.

The quadratic model adds to Equation 3.1 with t^2 and its interactions:

$$+ \beta_{16} t^2 + \beta_{17} (t^2 \times \text{Position}_i) + \beta_{18} (t^2 \times \text{Venue}_m)$$

The cubic model further adds t^3 and its corresponding interaction terms:

$$+ \beta_{19} t^3 + \beta_{20} (t^3 \times \text{Position}_i) + \beta_{21} (t^3 \times \text{Venue}_m)$$

While polynomial models provide some level of flexibility compared to linear models, they still impose a set curvature. That is, the same polynomial curve describes the second half deterioration window for all players. These linear and polynomial models were tested to quantify the improvement of increasing polynomial degrees and to find the point at which the parametric flexibility reaches its peak. This motivates the move to a non-parametric model approach. Model selection for both linear and the polynomial models were done with backward elimination which sequentially removes non-significant fixed effects.

3.5 Generalized Additive Models

While polynomial models allow some curvature in the deterioration trajectory, this may not adequately represent the complex pattern of fatigue seen in the exploratory analysis. Generalized additive mixed models (GAM) address this by replacing the parametric polynomial terms with penalized regression

splines [15]. These penalized regression splines are flexible smooth function that adapt to the data [15] which may be well suited for modeling where deterioration may have non-linearity or inflection points. This GAM takes the form:

$$\begin{aligned} \text{Deterioration}_{i,t+\ell} = & s(\text{Deterioration}_{i,t}) + s(t \mid \text{Position}_i) + s(t \mid \text{Venue}_m) + s(t \mid \text{DaysSince}_i) + s(t \mid \text{Peak}_i) \\ & + s(\text{CumHSD}_{i,t} \mid \text{Position}_i) + s(\text{CumHSD}_{i,t}) + s(\text{CumDist}_{i,t} \mid \text{Position}_i) + s(\text{CumDist}_{i,t}) \\ & + s(\text{DaysSince}_i) + s(\text{DaysIntoSeason}_m) + s(\text{Peak}_i) + \text{Position}_i + \text{Venue}_m \\ & + s(\text{athlete_id}, bs = \text{re}) + s(\text{match_id}, bs = \text{re}) + \epsilon_{i,t} \end{aligned} \quad (3.2)$$

where $s(\cdot)$ is a penalized regression spline and $s(\cdot \mid z)$ is a smooth of match minute t estimated separately for each level of a grouping variable z , allowing the deterioration trajectory to vary by position group, game state, venue, individual recovery time, and first half peak. The random effects for players and matches are included as penalized smooths with random effect.

The key advantage of the GAM over the polynomial models is that each smooth term find its own optimal shape from the data. For instance, the deterioration trajectory of a center forward playing a home game doesn't need to follow the same trajectory as a winger playing an away game. Two distributional families were investigated, Gaussian and scaled t . The Gaussian family was fitted as the standard baseline assumption. The bounded behavior of the deterioration definition, which has upper bounds of 1, can produce asymmetric residual distributions at high deterioration values that the heavier tails of the scaled t can better treat.

Instead of backward selection, variable selection in the GAM models were handled automatically with an added shrinkage penalty that changes the standard smoothing penalty with a second penalty that shrinks smoothing terms to zero, thus automatically excluding non-contributing predictors from the model.

4 Model Evaluation

4.1 Train Test Data Split

To evaluate predictive performance on unseen match data, the 35 available matches were split into a training set of 28 matches and a testing set of 7 matches (corresponding to a 80/20 split). The split was determined by searching through 500 random seeds and selecting the seed that produced the most balanced allocation across result distribution (win, draw, loss), match venue (home, away), season thirds (early, mid, late)), and mean fatigue score distribution. The resulting validation set comprises 3 wins, 3 draws, and 1 loss, with 4 home and 3 away fixtures drawn from across all thirds of the season. The training set contains the remaining 28 matches. Given the small number of available matches, test metrics carry sampling uncertainty and results should be interpreted with caution as discussed in Section 6.2.

4.2 Model Comparison

Table 2 presents the training and test metrics for all five models (linear, quadratic, cubic, GAM Gaussian, and GAM scaled t) across lag values $\ell = 1, 5, 10$ minutes. Models are compared on training AIC, training Root Mean Square Error (RMSE) and Mean Absolute Error (MAE) to check sample fit as well as test set RMSE and MAE to check predictive accuracy. Note that AIC of the scaled t GAM is not directly comparable to the other Gaussian specifications due to different likelihood functions. Model selection and predictive results are informed by these metrics.

We acknowledge that due to limited amount of data, our test set is a small set of seven matches which can introduce uncertainty into the reported test metrics which we use for model selection. For instance, the test RMSE was consistently slightly lower than training RMSE across all model specifications. This can be attributed to the lower variance of the test set. Therefore, differences in

test metrics should be interpreted cautiously. Ideally we would also have another test set of unseen matches, however, due to the limited number of matches available in the dataset, building multiple test sets was not achievable.

Lag	Model Type	Family	Train				Test	
			AIC	RMSE	MAE	Deviance Explained	RMSE	MAE
1	Linear	Gaussian	-6498	0.173	0.131	NA	0.163	0.126
	Quadratic	Gaussian	-6521	0.173	0.131	NA	0.163	0.127
	Cubic	Gaussian	-6535	0.173	0.131	NA	0.163	0.126
	GAM	Gaussian	-6816	0.169	0.129	73.2%	0.164	0.128
	GAM	Scaled t	-7210	0.170	0.129	62.9%	0.165	0.128
5	Linear	Gaussian	2975	0.290	0.224	NA	0.267	0.211
	Quadratic	Gaussian	2910	0.289	0.223	NA	0.270	0.213
	Cubic	Gaussian	2902	0.289	0.223	NA	0.270	0.213
	GAM	Gaussian	1383	0.255	0.197	39.7%	0.281	0.221
	GAM	Scaled t	1087	0.258	0.195	32.5%	0.282	0.221
10	Linear	Gaussian	2466	0.286	0.219	NA	0.267	0.208
	Quadratic	Gaussian	2442	0.285	0.219	NA	0.268	0.210
	Cubic	Gaussian	2414	0.285	0.219	NA	0.265	0.209
	GAM	Gaussian	1150	0.254	0.195	40.1%	0.278	0.220
	GAM	Scaled t	789	0.255	0.191	32.8%	0.274	0.216

Table 2: Model comparison across forecast lengths. Training metrics reflect in-sample fit on 28 matches. Test metrics reflect performance on 7 held-out matches used for model selection support.

4.3 Model Selection

When selecting a model for each lag value we consider test metrics, parsimony, and training fit. At lag $\ell = 1$, all five models showed nearly identical predictive ability from the test metrics with differences only as high as 0.001 between GAM Gaussian and the simpler linear and polynomial mixed-effect models. All five models were indistinguishable by test metrics so we picked the linear mixed-effect as our preferred model on grounds of parsimony and so to minimize risk of overfitting. At lag $\ell = 5$, the linear model had the best performing test metrics with RMSE of 0.2672, outperforming the next best, cubic model, by 0.003, it was therefore chosen as the preferred model. At lag $\ell = 10$, the cubic mixed-effect model had a slightly lower RMSE of 0.265 compared to the linear mixed-effect model with RMSE of 0.267. This difference is very small and the MAE of the linear mixed-effect model is the smallest at 0.208. So, the linear mixed-effect model was selected for consistency, parsimony, and performance.

4.4 Predictive Diagnostics

Diagnostic plots were presented for the selected linear mixed-effect model predictions evaluated on the test set of matches a lag lengths $\ell = 1, 5, 10$. Four diagnostics were examined: observed vs fitted for accuracy, residual distribution for bias, residuals vs fitted for heteroskedasticity, and residuals vs match minute from temporal consistency.

For the one minute lag model shown in 8, the observed vs fitted plot has a clear positive relationship with points clustering closely around the diagonal, indicating good predictive model. The residuals vs fitted plot shows an even spread around zero confirming the absence of significant heteroskedasticity. The residual distribution is symmetric and centered at zero indicating unbiased predictions, and the residuals vs match minute plot is flat across showing the consistency of predictions. These diagnostics indicate that the lag 1 linear mixed-effects model produces relatively well done one minute ahead predictions.

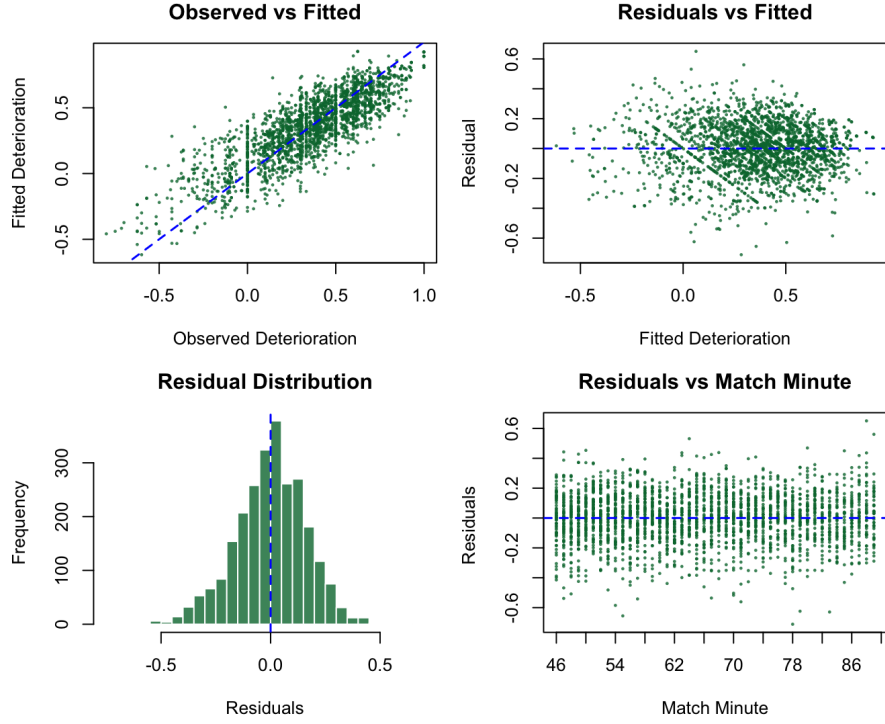


Figure 8: Predictive diagnostic plots for the selected linear mixed-effects model at lag $\ell = 1$ on the seven-match test set.

For the five minute lag model shown in Figure 9, the observed vs fitted plot shows a slight positive relationship but much more diffused than the one for lag 1 indicating that the model has weak predictive power. The residual distribution remains approximately centered at zero and the temporal plot of residuals vs match minute is flat indicating no bias or temporal degradation. These diagnostics reveal that lag 5 linear mixed-effects model cannot reliably distinguish between players who will be highly or moderately deteriorated five minutes ahead, limiting its utility for precise individual forecasting.

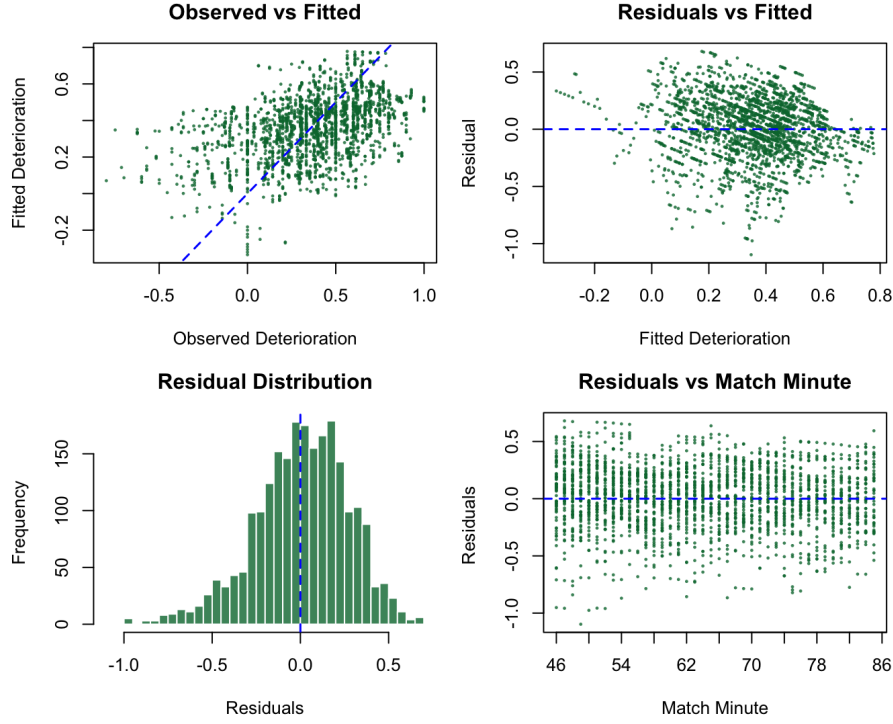


Figure 9: Predictive diagnostic plots for the selected linear mixed-effects model at lag $\ell = 5$ on the seven-match test set.

For the 10 minute lag model shown in 10, the observed vs fitted and residual vs fitted plots are similar to those of the lag 5 model, indicating poor predictive behavior. The residual distribution is also centered at zero and the residuals vs match minute shows a flat but slightly wider spread suggesting some uncertainty. The plots for lag 5 and lag 10 models are nearly identical and indicate that predictive accuracy plateaus beyond the five minute forecast with the current working linear mixed-effects model.

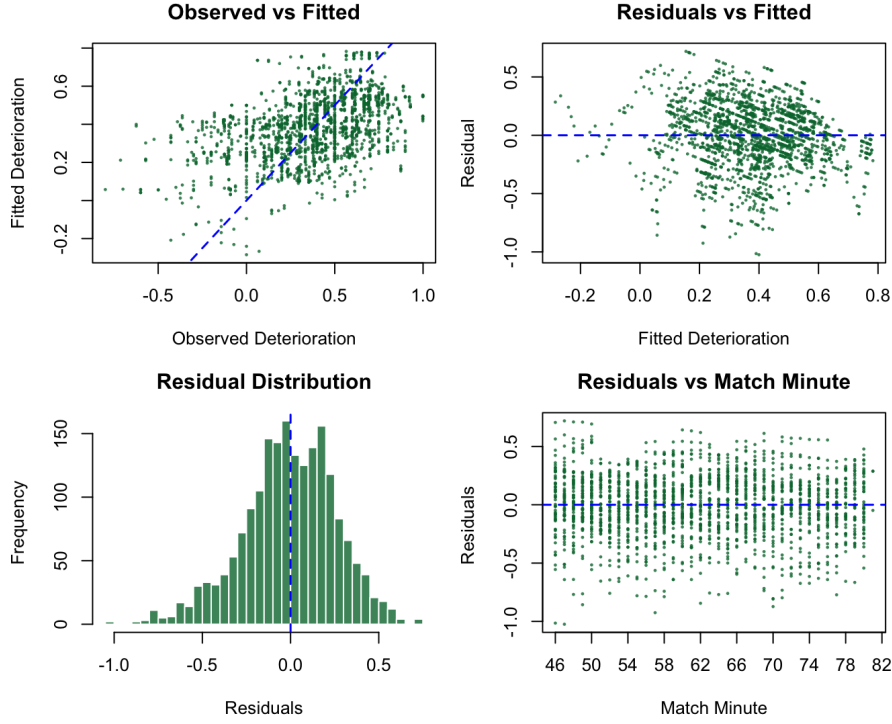


Figure 10: Predictive diagnostic plots for the selected linear mixed-effects model at lag $\ell = 10$ on the seven-match test set.

A central finding of the research so far is that deterioration can be reliably characterized in real-time and validated as a meaningful indicator of fatigue in professional football players as demonstrated in 2.4. However, the accurate prediction of future deterioration at longer lengths of five and ten minutes ahead is beyond the capability of the models developed in this paper. That is precise future deterioration values cannot be reliably predicted at these timescales with the available GPS-derived predictors. These limitations are acknowledged in the monitoring and application framework where those longer look ahead forecasts are treated as signals for identifying players at risk instead of precise future predictions. The inability to predict precise future deterioration values does not weaken the practical contribution of the monitoring capabilities that coaching staff can refer to.

5 Applications to Real-time Monitoring & Flagging Fatigued Players

In discussion with industry partners at Hibernian FC, the coaching staff seek to identify which players are currently or imminently the most fatigued. While we found in Section 4.4 that our models were not suboptimal for predictive use, we were motivated by an alternative approach to evaluating fatigue in football players in terms of ranking accuracy and real-time deterioration insights.

5.1 Ranking Accuracy

In ranking accuracy we explored what proportion of monitoring match minutes (using the test data) does our chosen linear mixed-effects model correctly flag the top deteriorated players from those on the pitch. The idea is that relative ordering of players is preserved even when the predicted values deviate from their true values (with the assumption that errors are not larger for more deteriorated players). Additionally, the near zero mean residuals observed in diagnostic analysis of Section 4.4 suggest that errors are approximately unbiased across the deterioration range which supports the use of ranking accuracy as a relevant evaluation. The lag 1 model (predicting deterioration for minute $t + 1$) correctly identified at least three of the four most deteriorated players at 85.7% of second half minutes. The flagging accuracy declined for longer lags as expected with the lag 5 model ($t + 5$) correctly identifying

at least three of the four at 62.4% and the lag 10 model ($t + 10$) correctly identifying at least three of the four at 66.9%. These results suggest that while the models lack strong precise prediction ability of future deterioration but have the ability to narrow down the field of concern to those players most at risk providing coaching staff with a reliable indication of which players warrant their attention during the second half.

5.2 Real-time Deterioration Insights

In line with discussion with industry partners at Hibernian FC on presenting current fatigued players in a match to coaching staff, we felt the deliverable of real-time deterioration insights would be especially valuable. The first insight deliverable was built with the intention of being used during half-time. This presents players who were the most deteriorated by the end of the first half to flag players entering the second half already showing signs of elevated fatigue. Figure 11 presents an example output from a sample match from the test set.

Hibernian FC Half Time Fatigue Report Match 4006418 Score 1-2 (Losing)

Rank	Player	Position	Deterioration
1	Warren O'Hora	CB	55.5%
2	Dylan Levitt	DM	50.0%
3	Jamie McGrath	AM	50.0%
4	Nicky Cadden	FB	40.7%
5	Jack Iredale	CB	25.0%

Figure 11: half-time fatigue report for sample test match presenting the five most deteriorated players at the end of the first half ranked by deterioration.

Another insight is a summary presenting the top 5 players deteriorated in that minute, what level of deterioration they have achieved, and a corresponding plot of their current second half deterioration trajectory. During a second half match minute, all players on the pitch are ranked by their current observed deterioration. No model is required for this, just an application of Equations 2.1, 2.2, and 2.3. Figure 12 presents an example output from a sample match from the test set.

Hibernian FC Fatigue Monitor Minute 68 Match 4006500 Score 2-1 (Winning)

Rank	Player	Position	Deterioration
1	Grant Campbell Hanle	CB	61.5%
2	Josh Mulligan	FB	61.5%
3	Jordan Obita	CB	40.0%
4	Daniel Barlasier	DM	30.7%
5	Martin Boyle	CF	18.2%

Figure 12: real-time fatigue monitoring output for a sample test match displaying the five most deteriorated players ranked by deterioration.

Both these insights, if built into deliverables that could be used live in match, would benefit the decision making and planning that coaching staff work through in a match by presenting which players to keep an eye on.

6 Discussion

6.1 Key Findings

This study developed and evaluated a real-time fatigue monitoring framework for professional football using GPS and accelerometer data from Hibernian FCs 2025/26 Scottish Premiership season. Three key findings emerged from this analysis.

First, deterioration defined as the decline in a players rolling high-intensity acceleration and deceleration output relative to first half peak was validated as a meaningful signal of within-match fatigue. Players who were substituted during the second half exhibited systematically higher deterioration than those who completed the full match, with broader literature establishing high-intensity accelerations and decelerations as among the most sensitive indicators fo fatigue [8].

Second, the model comparison revealed a consistent pattern across all three lags with the GAM Gaussian achieving the better training fits than all other models but the linear mixed-effects model producing the best or equivalent test performance by RMSE in all three lags. This shows that the better training fit of the GAMs additional flexibility captures match specific training patterns that do not generalize to unseen matches. This is consistent with findings in applied sports science modeling where simpler parametric models generalized well.

Third, whilst precise prediction of future deterioration at longer lags was found to be beyond the capability of the available data, the models demonstrated meaningful flagging accuracy. The results of the flagging accuracy indicate that while the models cannot provide precise forecasts of individual player deterioration, it can narrow down who to keep an eye on during the second half.

Overall, the monitoring framework developed had direct practical implications for the coaching staff at Hibernian FC. The real-time and half-time deterioration insights provides coaching staff with an objective and directly interpretable view into player fatigue and, in turn, who may be prioritized for substitution.

Of important note is that these outputs do not replace coaching judgment but rather work alongside it. Substitution decisions involve tactical considerations, injury assessment, and large amount of contextual factors that physical monitoring alone could not fully capture. This analysis is designed as decision support rather than a strict recommendation system. The validation of deterioration against observed substitution in Section 2.4 provides evidence that the metric captures something coaching staff are already implicitly responding to suggesting the framework can make this assessment explicit.

6.2 Limitations and Future Applications

Limitations have been discussed throughout this research. Two in particular should be acknowledged when interpreting findings and considering future research and applications.

6.2.1 Absence of Internal Load Metrics and Tactical Contamination of External Load Metrics

As discussed in Section 2, the model relies on exclusively external GPS and accelerometer-derived metrics as proxies for player fatigue. These external load variables capture mechanical demands but misses out on the internal physiological player states that make up fatigue. Metrics such as heart rate and heart rate variability could provide a more full picture of player fatigue state and have been previously found to be more sensitive indicators of physiological stress [7]. The consequence of the absence of internal metrics is that the models may fail to flag players who are physiologically fatigued but externally performing well, potentially underestimating the true extent of fatigue in some cases. Future research could investigate whether integrating internal load data alongside external load data improves the accuracy and interpretability of deterioration monitoring and predictions. Additionally,

deterioration measures are susceptible to tactical contamination that we cannot necessarily distinguish from the given data. This may potentially generate false positive high deterioration flags as HIA and HID can be driven by coaching instructions not just physical fatigue. Future work would explore methods for detangling tactical contributions, such as possession metrics or defensive shape.

6.2.2 Small Data Set

The data set provided is relatively small with only one Premiership season consisting of only 36 matches. This limits the splitting of the data into training and testing for the models as we were only able to feasibly split into one test set that was used for model comparison. Ideally you would have a training set, validation set to tune models, and a test set to run models on unseen data. Since we were unable to do this without risking the training of models. Additionally, a small data set split 80/20 for testing leaves us with a very small test set of just 7 matches which can create potential misleading results such as with variability of those matches.

7 Conclusion

Within this report, we have developed and validated a real-time within-match fatigue monitor for Hibernian FC using GPS and accelerometer data from the 2025/26 Scottish Premiership season. Deterioration defined as the decline in high-intensity explosive output relative to individual players first half peaks was established as a valid signal of physical fatigue. With this definition, a linear mixed-effects model was selected as the best performing model and was applied for future fatigue predictions. It was found that precise predictions of future deterioration at longer forecasts (five and ten minutes) was beyond the capability of the available metrics. However, the models retained the ability to identify the most physically deteriorated players with meaningful flagging accuracy. The resulting fatigue monitoring set up with real-time and half-time deterioration reports provides coaching staff with interpretable evidence to support substitution decision making.

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Appendices

A Data

A.1 Data Structure and Sources

Match and player data was provided by Hibernian FC covering their 2025/26 season in the Scottish Premiership, this spans 35 matches. Players wear tracking vests containing 10 Hz GPS receivers that record each players position ten times per second and obtain distance and velocity metrics. 100 Hz accelerometers record players movements in three directions 100 times per second. From these measurements five metrics are computed per player per minute: total distance covered (m), high speed distance covered (m, movements greater than 19.8 km/h), maximum velocity (m/s), high-intensity accelerations and decelerations (counts, changes in velocity $\pm 3\text{m/s}^2$). Match metrics including game state (winning, drawing, or losing) and real-time scores (goals for and against) were also recorded for each minute of play. Note that stoppage time in both halves is excluded from the analysis. Fixture information such as match date, kick-off time, venue (home/away) for all matches was also provided. Player identifiers including name and position were also provided.

A.2 Preprocessing

The data set includes 130 of 484 player match appearances beginning after minute 45, corresponding to substituted players entering the match. Including these appearances in a fatigue analysis introduces bias as substitutes enter the game fresh with no accumulated match load which may deflate the fatigue signal. Therefore the appearance type *starter* is defined as any player whose first recorded minute is minute 0 of the first half. All analyses use the sample of starters only. Additional variables are derived from the raw data prior to modeling:

- **Days Since Last Match** The number of days between each player’s current and most recent previous appearance in the season, computed for each individual player.
- **Days Into Season** The number of days since the first match of the 25/26 season computes at the match level used as a proxy for cumulative seasonal load as mentioned by Industry partners at Hibernian FC.
- **Rolling Metrics** A right-aligned rolling 10 minute window was applied to high-intensity accelerations (HIA), high-intensity decelerations (HID), and high speed distance (HSD) for each player within each match.
- **Fatigue Score** The fatigue score at minute t is defined as the rolling window 10 minute sum of HIA and HID events. Motivation for this selection of HIA and HID over other available metrics is discussed in Section 2.
- **Deterioration** The degree to which a players current fatigue score has declined relative to their individual first half peak (90th percentile of players fatigue score). Further explanation, calculations, and validation can be found in Sections 2